

YASH RUPANI

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Master's graduate with **2+ years of enterprise experience** building scalable **ETL pipelines** and real-time data architectures on **AWS** and **GCP**. Expert in **Apache Spark**, **Kafka**, and AI-driven data systems (**RAG**, **Vector DBs**). Proven track record of reducing infrastructure costs and optimizing data retrieval latency.

Education:

Master of Engineering in Computer Science, Oregon State University, USA

Sep 2023–Dec 2025

Bachelor of Technology in Electrical Engineering, PDEU, India

Aug 2017–Jun 2021

Technical Skills:

- **Languages:** Python, SQL, Bash, C++.
- **Data Engineering:** Apache Spark, PySpark, Kafka, Airflow, Snowflake, dbt, Pandas, NumPy.
- **Cloud & DevOps:** AWS (S3, Lambda, Glue, EMR, Redshift), GCP (BigQuery, Firestore), Docker, Kubernetes.
- **AI & Databases:** Vector Databases (FAISS), RAG Architectures, PostgreSQL, NoSQL, Dimensional Modeling.

Professional Experience:

Oregon State University (OR, USA)

Oct 2025–Dec 2025

Research Assistant – Data Engineering Focus

- Engineered an end-to-end autonomous **ETL pipeline** for unstructured multimedia data, leveraging **Python**-based preprocessing to eliminate **40%** of manual data preparation for **Agentic AI** workflows.
- Refined backend data structures and indexing strategies for **AI Agents**, slashing retrieval latency by **25%** and boosting real-time inference performance.
- Developed robust data validation frameworks to ensure **100% data integrity** while enabling scalable ingestion of **terabyte-scale** datasets.

GrantAide (CA, USA)

Jul 2025–Sep 2025

AI Agent Graph RAG / Machine Learning Scientist Intern

- Architected scalable backend systems harnessing **Flask** and **Google Firestore**, restructuring database access patterns to reduce API latency by **30%** during peak usage.
- Orchestrated **FAISS-based vector search** pipelines to enable semantic retrieval, increasing search relevance by **40%** compared to traditional keyword matching.
- Managed multi-cloud deployments across **AWS** and **GCP**, achieving **99.9% uptime** while reducing cross-environment latency by **25%**.

Infosys (Maharashtra, India)

Jun 2021–Jun 2023

Sr. Systems Engineer (Data Engineering)

- Spearheaded the modernization of enterprise **data quality pipelines**, automating workflows to eliminate **10+ hours/week** of manual intervention.
- Designed high-performance **ETL workflows** using **Apache Spark** to process large-scale system monitoring data, compressing runtime by **40%** per cycle.
- Constructed dynamic dashboards from semi-structured logs, replacing manual analysis with **automated monitoring** solutions.

Projects:

Crypto Sentinel: Real-Time Market Monitor (Next.js, FastAPI, Spark, Kafka, MongoDB) [[Live Demo](#)]

- Engineered a real-time **streaming platform** using **FastAPI** and **Kafka** to ingest market data, reducing data latency by **40%** compared to traditional polling methods.
- Built a robust **WebSocket** alert system processing **500+ events/second**, triggering instant notifications when asset prices breach user-defined thresholds.

Shop Pulse: Real-Time E-Commerce Lakehouse (Spark, Kafka, Python, Docker) [[Live Demo](#)]

- Architected a **Lakehouse pipeline** leveraging **Spark Streaming** and **Kafka**, achieving **sub-second latency** for live data visualization.
- Deployed a fault-tolerant storage layer using **partitioned Parquet** files and **Docker** containerization, optimizing query performance for downstream analytics.